Measurement Based Quantum Computation

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In this paper, I will provide an brief summary of measurement based quantum computation model following the construction of Raussendorf, Browne, and Briegel in 2008. This measurement based model simulates unitary quantum logical network by preparing a string of entangled quantum qubits, known as the cluster state, and perform a sequence of measurement to teleport and computation of some input information. Concatenating cluster construction for single qubit quantum gates immeditately leads to the construction of quantum logical network. An analysis on temporal, operational, and spacial resource consumption also shows that measurement based quantum computation provides an efficient universal construction of quantum computation scheme.

I. INTRODUCTION

The most used model to explain quantum computation is traditionally quantum circuits. Quantum circuits are commonly used to design computation schemes such as quantum error correction and Shor's algorithm. Experiments are also designed to realize the quantum circuits as a logical network of quantum logical gates. For the rest of the paper, quantum circuit and quantum logical network will be used synonymously.

An alternative way of realizing quantum computation is through measurement based quantum computation. Robert Raussendorf and Hans Briegel first introduce such scheme in their paper from 2000. Instead of using gates and wires, it proposes a mechanism which realize the computation and transportation of information through a series of specifically designed and adaptive single qubit measurements on a highly entangled qubit cluster state. The entire resource of the computation is provided initially in this entangled cluster. Since measurement would destroy the cluster state, this model is also known as one way quantum computation. This model realizes all the necessary unitary computation of single quantum gate computation and can be concatenated to perform composite computation. As I will show below, it is an effective universal scheme of quantum computation, and can be thought as a simulator of quantum circuits.

I.1. Quantum Teleportation

In this section, I will give a brief summary of quantum teleportation as a motivation of measurement based model. In the simplest case, quantum teleportation is realized on a 3-qubit state. Assume that Alice holds two qubit and Bob holds one, and Alice wants to teleport one of her qubit, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ to Bob.

First, Alice and Bob entangle their qubits (that is not the one waiting to be teleported) to a maximally entangled state. Then, the three qubit state becomes

$$|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{2}}\left(|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle\right)$$

Then, Alice performs a unitary transformation on her two qubits $U = (H \otimes \mathbb{I})CNOT$, the resulting state is

$$\begin{aligned}|\Psi'\rangle = |00\rangle|\psi\rangle + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \\ + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)\end{aligned}$$

The input qubit has successfully transported to Bob's Hilbert space, with a distortion. However, this distortion can be corrected based on the measurement on Alice's Hilbert space.

- If Alice has measurement output $|00\rangle$, the qubit is not altered, hence we apply $\mathbb{I}$ on Bob's qubit.
- If Alice's measurement output is $|01\rangle$, the qubit experiences a bit flip, hence we apply a X gate on Bob's qubit.
- If Alice's measurement output is $|10\rangle$, the qubit experiences a π phase shift, so we apply a Z gate on Bob's qubit.
- If Alice's measurement output is $|11\rangle$, the qubit experiences both a bit flip and a π phase shift. Hence, we apply a ZX gate on Bob's qubit.

Combing these three steps, we successfully transport the qubit from Alice's Hilbert space to Bob's Hilbert space without any error.

Similarly, with alteration of the size of qubit chain size, entanglement methods, and other details, both a teleportation and unitary transformation can be done over this qubit chain with specified measurement pattern. The paper will be organized as follows: in Sec.II, I will present how to prepare a chain of qubits, known as a cluster state, where the computation happens upon. In Sec.III, I will give the example of construction CNOT gates and $SU(2)$ rotations using this scheme. Finally, in Sec.IV, I will show how a general quantum circuit can be realized as an overall measurement pattern by concatenating single quantum gate clusters.

II. CLUSTER STATE

II.1. Definition of Cluster State

To allow the operation of quantum circuit on a lattice of qubits, a cluster state will be needed. Consider an

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n dimensional lattice of qubits. An n-qubit cluster $\mathcal{C}$ is a connected subset of the lattice containing n qubits. Then, the cluster state $|\phi_{\{\kappa\}}\rangle_{\mathcal{C}}$ is direct product of the n-qubit state following:

$$K^{(a)} = \sigma_x^{(a)} \bigotimes_{b\in nbgh(a)} \sigma_z^{(b)}$$
$$K^{(a)} |\phi_{\{\kappa\}}\rangle_{\mathcal{C}} = (-1)^{\kappa_a} |\phi_{\{\kappa\}}\rangle_{\mathcal{C}} \quad \forall a \in \mathcal{C} \tag{1}$$

where $nbgh(a)$ is the neighborhood of a, namely the set of sites that are connected to site a, and $\kappa_a \in \{0,1\}$. Further, the set $\{\kappa_a | a \in \mathcal{C}\}$ specifies the cluster state $|\phi_{\{\kappa\}}\rangle_{\mathcal{C}}$. If two cluster states has κ, κ' different by at least one element, then the two cluster states are orthogonal to each other. Hence, the set of $\{|\phi_{\{\kappa\}}\rangle_{\mathcal{C}} | \{\kappa\} \in \{0,1\}^{|\mathcal{C}|}\}$ forms an orthogonal basis of the $2^{|\mathcal{C}|}$ dimensional Hilbert space.

II.2. Preparation

The first step of create a cluster state on a $\mathcal{C}_n$ cluster is to prepare a product state $|+\rangle_{\mathcal{C}} = \bigotimes_{a\in\mathcal{C}} |+\rangle_a$. Then, apply a unitary transformation

$$S^{(\mathcal{C})} = \prod_{a,b\in\mathcal{C}|b\in nbgh(a)} S^{ab} \tag{2}$$
$$S^{ab} = \frac{1}{2}\left(1 + \sigma_z^{(a)} + \sigma_z^{(b)} - \sigma_z^{(a)} \otimes \sigma_z^{(b)}\right)$$

This unitary transformation impose a phase π on the state $|1\rangle_a \otimes |1\rangle_b$, and keeps the rest of the states $|0\rangle_a \otimes |0\rangle_b$, $|0\rangle_a \otimes |1\rangle_b$, $|1\rangle_a \otimes |0\rangle_b$ invariant.

Following this construction, we can write out the cluster state on 2, 3, 4 qubit cluster.

$$|\phi_{\{\kappa\}}\rangle_{\mathcal{C}2} = \frac{1}{\sqrt{2}}\left(|0\rangle_1 |+\rangle_2 + |1\rangle_1 |-\rangle_2\right)$$
$$|\phi_{\{\kappa\}}\rangle_{\mathcal{C}3} = \frac{1}{\sqrt{2}}\left(|+\rangle_1 |0\rangle_2 |+\rangle_3 + |-\rangle_1 |1\rangle_2 |-\rangle_3\right)$$
$$|\phi_{\{\kappa\}}\rangle_{\mathcal{C}4} = \frac{1}{2}(|+\rangle_1 |0\rangle_2 |+\rangle_3 |0\rangle_4 + |+\rangle_1 |0\rangle_2 |-\rangle_3 |1\rangle_4 + |-\rangle_1 |1\rangle_2 |-\rangle_3 |0\rangle_4 + |-\rangle_1 |1\rangle_2 |+\rangle_3 |1\rangle_4)$$

III. CONSTRUCTION OF SINGLE QUANTUM STATE

III.1. CNOT

In this construction, a CNOT gate is realized through a 15 qubit cluster, following the below steps:

- Prepare the initial state $|\psi_{in}\rangle_{1,9} \left(\bigotimes_{i\in\mathcal{C}_{15\setminus\{1,9\}}} |+\rangle_i\right)$
- Entangle the 15 qubit cluster with the unitary transformation $S^{(\mathcal{C}_{15})}$

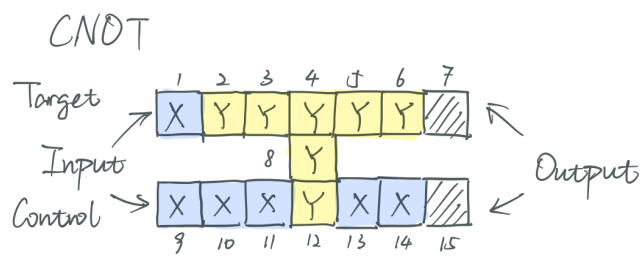

FIG. 1. Each box indicates a qubit site. The figure provides the measurement pattern on 15 qubit cluster to realize a CNOT gate.

- One qubit measurement is performed on every qubit in the cluster except for the output qubit(7, 15). The measurement pattern is visualized in Fig.1.
  - σ_x measurement: qubit 1, 9, 10, 11, 13, 14
  - σ_y measurement: qubit 2, 3, 4, 5, 6, 8, 12
  - The measured qubits would be destroyed and resulting gate is $U'_{CNOT} = U_{\Sigma,CNOT} CNOT(c,t)$
  - $U_{\Sigma,CNOT} = \sigma_x^{(c)^{\gamma_x^{(c)}}} \sigma_x^{(t)^{\gamma_x^{(t)}}} \sigma_z^{(c)^{\gamma_z^{(c)}}} \sigma_z^{(t)^{\gamma_z^{(t)}}}$ with

$$\gamma_x^{(c)} = s_2 + s_3 + s_5 + s_6$$
$$\gamma_x^{(t)} = s_2 + s_3 + s_8 + s_{10} + s_{12} + s_{14}$$
$$\gamma_z^{(c)} = s_1 + s_3 + s_4 + s_5 + s_8 + s_9 + s_{11} + 1$$
$$\gamma_z^{(t)} = s_9 + s_{11} + s_{13}$$

III.2. $SU(2)$ Rotation

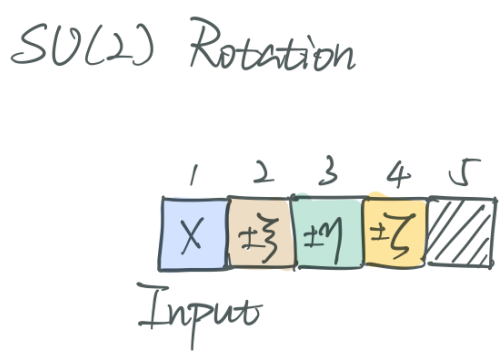

FIG. 2. The figure provides the measurement pattern on 5 qubit cluster to realize an arbitrary SU(2) rotation.

Similarly, a general $SU(2)$ rotation can be realized on a 5 qubit cluster. For any $U \in SU(2)$ on a qubit, it can be decomposed into the form $U = U_X[\zeta]U_Z[\eta]U_X[\xi]$, correspond to Euler rotation and its three Euler angles. The procedure is as follows:

- Prepare the state $|\Psi\rangle_{\mathcal{C}_5} = |\psi_{in}\rangle_1 \left(\bigotimes_{i=2}^{5} |+\rangle_i\right)$
- Create the cluster state by performing the unitary operation $S^{(\mathcal{C}_5)}$
- Measure the first 4 qubits in the following order. Each measurement correspond to a step of Euler rotation
  - Measure qubit 1 in $\mathcal{B}_1(0)$.
  - Measure qubit 2 in $\mathcal{B}_2(-\xi(-1)^{s_1})$.
  - Measure qubit 3 in $\mathcal{B}_3(-\eta(-1)^{s_2})$.
  - Measure qubit 4 in $\mathcal{B}_4(-\zeta(-1)^{s_2+s_3})$.

  With $\mathcal{B}_j(\varphi_j) = \left\{\frac{|0\rangle_i + e^{i\varphi}|1\rangle_j}{\sqrt{2}}, \frac{|0\rangle_i - e^{i\varphi}|1\rangle_j}{\sqrt{2}}\right\}$
- Then, we have performed a rotation $U'_{Rot} = U_{\Sigma,Rot} U_{Rot}[\eta, \xi, \zeta]$, with the byproduct operator $U_{\Sigma,Rot} = \sigma_x^{s_2+s_4} \sigma_z^{s_1+s_3}$

III.3. Hadamard and $\pi/2$-Phase Gate

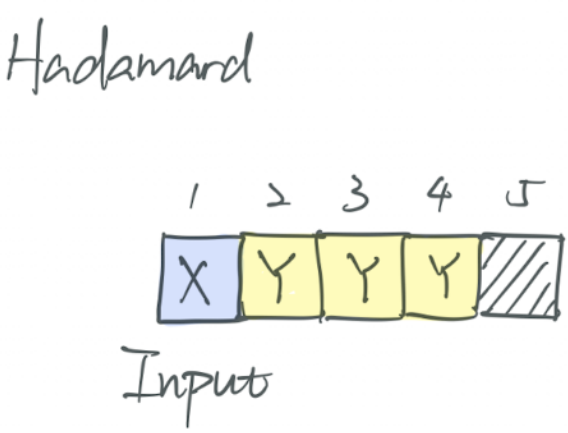

FIG. 3. The figure provides the measurement pattern on 5 qubit cluster to realize an arbitrary Hadamard Gate.

In the previous section, I described the scheme to construct the measurement pattern for a general $SU(2)$ rotation. The measurement must be performed in the order specified. However, there is a special class of single qubit gates where the measurement can be performed simultaneously. In this section, I will give two examples of such gates, Hadamard gate and $\pi/2$-phase gate.

Both gates are constructed on a 5-qubit cluster.

- Prepare the 5-qubit cluster state
- Measurement of the first 4 qubits can be performed simultaneously
  - Hadamard Gate: $\sigma_x^{(1)}$, $\sigma_y^{(2)}$, $\sigma_y^{(3)}$, $\sigma_y^{(4)}$, $U_{\Sigma,H} = \sigma_x^{s_1+s_3+s_4} \sigma_z^{s_2+s_3}$
  - $\pi/2$-phase Gate: $\sigma_x^{(1)}$, $\sigma_x^{(2)}$, $\sigma_y^{(3)}$, $\sigma_x^{(4)}$, $U_{\Sigma,H} = \sigma_x^{s_2+s_4} \sigma_z^{s_1+s_2+s_3+1}$.

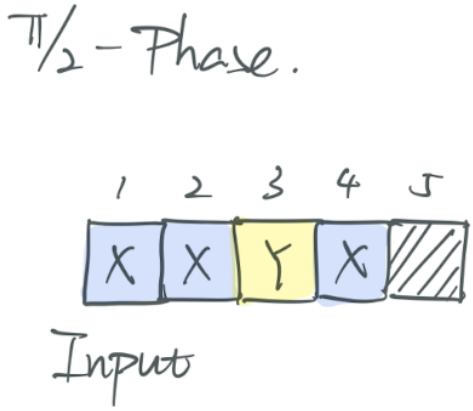

FIG. 4. The figure provides the measurement pattern on 5 qubit cluster to realize an arbitrary $\pi/2$-phase gate.

IV. REALIZING QUANTUM CIRCUIT

In this section, I will show how the measurement pattern indeed realize a quantum circuit and is equivalent as a chain a quantum gates one by one.

For any form of procedure, the cluster needed $\mathcal{C}(g)$ can be decomposed into three parts, an input cluster $\mathcal{C}_I(g)$, a body cluster $\mathcal{C}_M(g)$, and an output cluster $\mathcal{C}_O(g)$ with the properties

$$\begin{aligned} C_I(g) \cup C_M(g) \cup C_O(g) &= C(g) \\ C_I(g) \cap C_M(g) &= \emptyset \\ C_I(g) \cap C_O(g) &= \emptyset \\ C_M(g) \cap C_O(g) &= \emptyset \end{aligned}$$

Hence, to realize a general single gate we follow the scheme:

- Start with the state $|\psi_{in}\rangle_{1,9} \left(\bigotimes_{i \in \mathcal{C}_M \cup \mathcal{C}_O} |+\rangle_i\right)$
- Prepare the cluster state by applying $S^{(\mathcal{C}(g))}$
- Measure the qubits in the cluster $\mathcal{C}_I(g) \cup \mathcal{C}_M(g)$ with some projector

$$P^{(\mathcal{C}_I(g) \cup \mathcal{C}_M(g))} = \bigotimes_{k \in \mathcal{C}_I(g) \cup \mathcal{C}_M(g)} \frac{1 + (-1)^{s_k} \vec{r_k} \cdot \vec{\sigma}_k}{2}$$

  where s_k is some binary value based on the previous measurement outcomes, and $\vec{r_k}$ is a unit vector in the x-y plane.

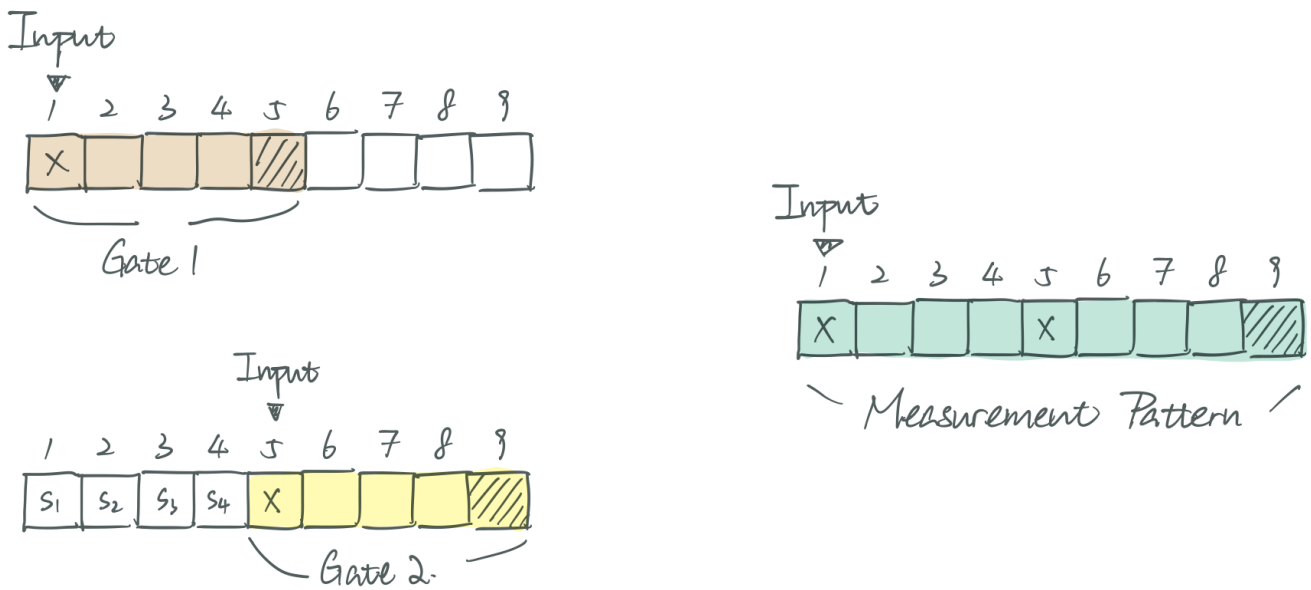

FIG. 5.

Then, the measurement output is

$$
\begin{aligned}
|\Psi\rangle &= P^{(\mathcal{C}_I(g)\cup\mathcal{C}_M(g))} S^{(\mathcal{C}(g))} |\psi_{in}\rangle_{1,9} \left(\bigotimes_{k\in\mathcal{C}_M\cup\mathcal{C}_O} |+\rangle_i \right) \\
&= \left(\bigotimes_{k\in\mathcal{C}_I(g)\cup\mathcal{C}_M} |s_k\rangle_{k,\vec{r_k}} \right) \otimes |\psi_{out}\rangle \\
&= \left(\bigotimes_{k\in\mathcal{C}_I(g)\cup\mathcal{C}_M} |s_k\rangle_{k,\vec{r_k}} \right) \otimes U_{\Sigma,g} U_g |\psi_{in}\rangle
\end{aligned}
$$

with $U_{\Sigma,g} = \bigotimes_{i=1}^{n} \left(\sigma_x^i\right)^{x_i} \left(\sigma_z^i\right)^{z_i}$. $\sigma_{x/z}^i$ denotes Pauli matrices acting on the i-th qubit in the output cluster, and x_i and z_i are binary values based on the previous measurement outcomes.

The entanglement operator commutes with single qubit measurements, as both of them act on individual qubit. Therefore, performing the scheme above individually for each quantum gate is equivalent to performing a global entanglement and then measurement only scheme. This statement can be shown in a simple consideration of a two quantum gate circuit.

Consider the cluster $\mathcal{C}$ where the two gates are performed on. Namely, the quantum circuit is a composite of two gates $g = g_1 \circ g_2$ is realized on a cluster $\mathcal{C}(g) = \mathcal{C}(g_1) \cup \mathcal{C}(g_2)$. Then, the following properties must be true:

$$
\begin{aligned}
\mathcal{C}_I(g) &= \mathcal{C}_I(g_1) \cup (\mathcal{C}_I(g_2) \setminus \mathcal{C}_O(g_1)) \\
\mathcal{C}_I(g) &= \mathcal{C}_O(g_1) \cup (\mathcal{C}_O(g_2) \setminus \mathcal{C}_I(g_1))
\end{aligned}
$$

Now, we perform the single quantum gate described before on $\mathcal{C}(g_1)$ with S_1 and P_1, then perform the same scheme on S_2 and P_2. Explicitly,

$$
\begin{aligned}
S_1 &= \prod_{a,b\in\mathcal{C}(g_1)|b\in nbgh(a)} S^{ab} \\
S_2 &= \prod_{a,b\in\mathcal{C}(g_2)|b\in nbgh(a)} S^{ab} \\
P_1 &= \bigotimes_{k\in\mathcal{C}_I(g_1)\cup\mathcal{C}_M(g_1)} \frac{1+(-1)^{s_k}\vec{r_k}\cdot\vec{\sigma}_k}{2} \\
P_2 &= \bigotimes_{k\in\mathcal{C}_I(g_2)\cup\mathcal{C}_M(g_2)} \frac{1+(-1)^{s_k}\vec{r_k}\cdot\vec{\sigma}_k}{2}
\end{aligned}
$$

The resulting cluster state is

$$
|\Psi\rangle_{out} = P_2 S_2 P_1 S_1 |\Psi\rangle_{in}
$$

Now, each qubit where the measurement has been performed on are destroyed. Hence, the condition $(\mathcal{C}_I(g_1) \cup \mathcal{C}_M(g_1)) \cap \mathcal{C}(g_2) = \emptyset$ must be true. It immediately follows that $P_2 S_2 P_1 S_1 = P_2 P_1 S_2 S_1$. Hence, we can define that

$$
\begin{aligned}
P^{\mathcal{C}_I(g)\cup\mathcal{C}_M(g)} &= P_2 P_1 \\
S^{\mathcal{C}(g)} &= S_2 S_1 \\
|\Psi\rangle_{out} &= P^{\mathcal{C}_I(g)\cup\mathcal{C}_M(g)} S^{\mathcal{C}(g)} |\Psi\rangle_{in}
\end{aligned}
$$

Therefore, without the loss of generality, we can show through induction that performing a local one qubit scheme is equivalent to performing a global unitary transformation then measurement-only scheme.

V. CONCLUSION

In this paper, I have summarized the measurement based quantum computation scheme. Motivated from

quantum teleportation, measurement on a string of qubits can both propagate and perform unitary transformation on input information. This requires proper entanglement of the qubits to form a cluster state using some unitary transformation described in Sec.II. Then, with special measurement pattern, whether measurement must be performed in order or simultaneously, quantum gates can be realized on this cluster, with a byproduct that can be corrected based on the measurement result. Finally, concatenation of a network of quantum gates is simply overlapping the cluster for individual quantum gate, which has been justified in Sec.IV.

In order to see the measurement based quantum computation provides for a universal model of quantum computing, its resource consumption must be reasonable. Hence, an analysis on temporal, operational, and spacial resources as well as the comparison to their quantum circuit construction counterparts are required.

Starting from the temporal resources, the temporal resource consumption is measured by finding the minimum measurement rounds to which the measurement can be compacted. For a CNOT gate on measurement based quantum computation, only one measurement is needed. For a general rotation, three measurements are needed. Combing those two aspects, the following inequality is established

$$T \leq 3T_{qc}$$

Then, we can further analyze for spacial and operational resource consumption. Consider a general cluster of height h and width w, so that the spacial resource consumption is $S = w \times h$. As can be observed similar in CNOT gate case, the height is to construct different wires, hence the height is $h = 4S_{gc} - 3$. Further, the vertically on at least one of the wire, there must be one quantum gate, hence the width is $w = 6S_{qc}T_{qc} + 1$. Since each measurement can contain at most one quantum gate,

$$S \leq 24S_{qc}^2 T_{qc}$$

Further, for since only single qubit measurements are performed, the operational resources needed must be less than spacial resources $O \leq S$. Also, from a quantum circuit, by definition $O_{qc} \leq S_{qc}T_{qc}$

$$S \leq 24O_{qc}S_{qc}$$
$$S \leq 24O_{qc}S_{qc}$$

Hence, all temporal, operational, and spacial resources are polynomial as compare to the quantum circuit construction. Hence, measurement based quantum computation has a similar efficiency as the quantum circuit model and is indeed an efficient computation scheme.